DT081A - Signal and Image Processing
Exercise 2 Solutions

Question 1. A continuous-time signal $x_c(t) = \cos(2\pi f_0 t)$, where $f_0 = 1$ kHz, is to be sampled and reconstructed.

(a) What is the minimum sampling frequency f_s required to sample $x_c(t)$ without aliasing?

(b) If the signal is sampled at $f_s = 1.5$ kHz, determine the frequency of the aliased component that appears in the reconstructed signal.

(c) Sketch the spectrum of the sampled signal when sampled at $f_s = 1.5$ kHz

Answer:

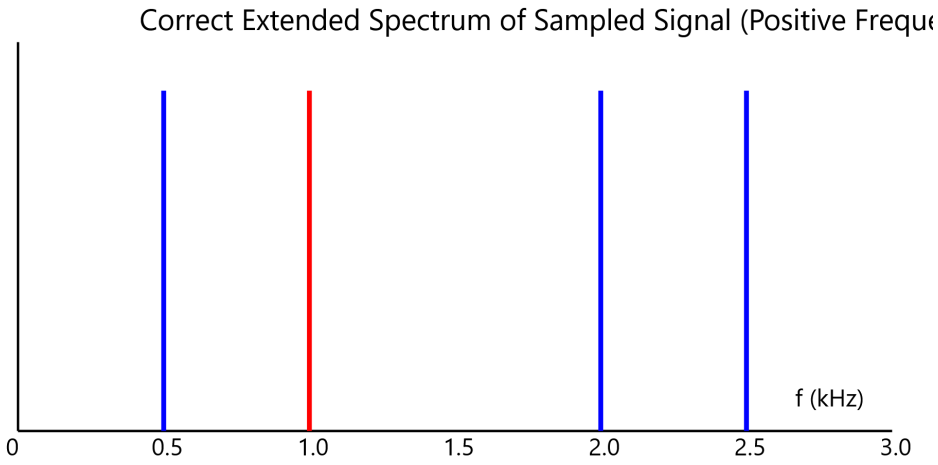
(a) The minimum sampling frequency required to avoid aliasing is given by the Nyquist-Shannon Sampling Theorem (NSST) (Lecture 2 Slide 21), which is twice the highest frequency component in the signal. For $x_c(t) = \cos(2\pi f_0 t)$ with $f_0 = 1$ kHz, the minimum sampling frequency is:

$$f_s \geq 2f_0 = 2 \cdot 1 \text{ kHz} = 2 \text{ kHz} \tag{1}$$

(b) When sampling at $f_s = 1.5$ kHz, which is below the rate given by the NSST, aliasing occurs. The aliased frequencies can be calculated as (Lecture 2 Slide 22):

$$F_{\text{alias}} = \{|1 - z1.5| \text{ kHz} \mid z \in \mathbb{Z}\} \tag{2}$$

(c) The spectrum of the sampled signal at $f_s = 1.5$ kHz would look like this:



Question 2. Given the signal $x(t) = e^{-at}u(t)$, where $u(t)$ is the unit step function and $a > 0$:

(a) Find the Fourier Transform $X(f)$ of $x(t)$.

(b) Using the time-shifting property, find the Fourier Transform $Y(f)$ of $y(t) = x(t - t_0)$, where $t_0 > 0$.

(c) Explain how the magnitude and phase of $Y(f)$ relate to those of $X(f)$.

Answer:

(a) The Fourier Transform of $x(t) = e^{-at}u(t)$ is:

$$X(f) = \int_{-\infty}^{\infty} e^{-at}u(t)e^{-j2\pi ft}dt \tag{3}$$

$$= \int_{0}^{\infty} e^{-at}e^{-j2\pi ft}dt \tag{4}$$

$$= \frac{1}{a + j2\pi f}. \tag{5}$$

(b) The time-shifting property states that if $x(t)$ has a Fourier Transform $X(f)$, then $x(t - t_0)$ has a Fourier Transform $e^{-j2\pi ft_0}X(f)$. Therefore:

$$Y(f) = e^{-j2\pi ft_0}X(f) \tag{6}$$

$$= e^{-j2\pi ft_0}\frac{1}{a + j2\pi f} \tag{7}$$

(c) The relationship between $Y(f)$ and $X(f)$ is:

Magnitude: $|Y(f)| = |X(f)|$ because $|e^{-j2\pi ft_0}| = 1$

Phase: $\angle Y(f) = \angle X(f) - 2\pi ft_0$, because the multiplication of $e^{-j2\pi ft_0}$ simply rotates $X(f)$ by $-2\pi f$.

Question 3. Given two signals:

$$x(t) = \text{rect}\left(\frac{t}{T}\right), \text{ a rectangular pulse of width } T,$$

$$h(t) = \text{sinc}\left(\frac{t}{T}\right), \text{ where } \text{sinc}(t) = \frac{\sin(\pi t)}{\pi t}.$$

(a) Compute the convolution $y(t) = x(t) * h(t)$.

(b) Explain the result using Fourier Transform properties and convolution.

Answer:

(a) The convolution is computed as:

$$y(t) = \int_{-\infty}^{\infty} \text{rect}\left(\frac{\tau}{T}\right) \text{sinc}\left(\frac{t-\tau}{T}\right) d\tau \tag{8}$$

$$= \int_{-T/2}^{T/2} \text{sinc}\left(\frac{t-\tau}{T}\right) d\tau \tag{9}$$

$$= T \int_{\frac{t}{T}-\frac{1}{2}}^{\frac{t}{T}+\frac{1}{2}} \text{sinc}(v) dv \quad (\text{where } v = \frac{t-\tau}{T}) \tag{10}$$

$$= T \int_{\frac{t}{T}-\frac{1}{2}}^{\frac{t}{T}+\frac{1}{2}} \text{sinc}(v) dv \tag{11}$$

$$= T \left(\text{Si}\left(\frac{t}{T}+\frac{1}{2}\right) - \text{Si}\left(\frac{t}{T}-\frac{1}{2}\right)\right) \tag{12}$$

(b) Note that the Fourier transformation of a rect is a sinc function and vice versa.

The convolution theorem states that convolution in the time domain is equivalent to multiplication in the frequency domain Using the individual transforms

$$X(f) = T \cdot \text{sinc}(fT) \tag{13}$$

$$H(f) = T \cdot \text{rect}(fT), \tag{14}$$

the convolution in the frequency domain is:

$$Y(f) = X(f) \cdot H(f) \tag{15}$$

$$= T^2 \cdot \text{sinc}(fT) \cdot \text{rect}(fT). \tag{16}$$

This is equivalent to $T^2 \cdot \text{sinc}(fT)$ within the interval $[-\frac{1}{2T}, \frac{1}{2T}]$ and zero elsewhere, due to the rectangular function.

The product $T^2 \cdot \text{sinc}(fT)$ in the frequency domain approximately corresponds to a scaled triangular function in the time domain. The rectangular function in frequency limits the bandwidth to $[-\frac{1}{2T}, \frac{1}{2T}]$, which results in the smoothing effect we see in the time domain.

In essence, the convolution has acted as a low-pass filter, smoothing out the sharp edges of the original rectangular pulse. This is evident in both the frequency domain analysis (multiplication by $\text{rect}(fT)$) and the time domain result (difference of sine integrals).

Question 4. An analog signal $x_c(t)$ has a frequency spectrum non-zero only between -3 kHz and 3 kHz.

(a) What is the minimum sampling frequency f_s required to sample $x_c(t)$ without aliasing?

(b) If the signal is sampled at $f_s = 4$ kHz, describe and sketch the resulting spectrum $X_s(f)$ of the sampled signal.

(c) Propose a method to prevent aliasing when sampling at $f_s = 4$ kHz.

Answer:

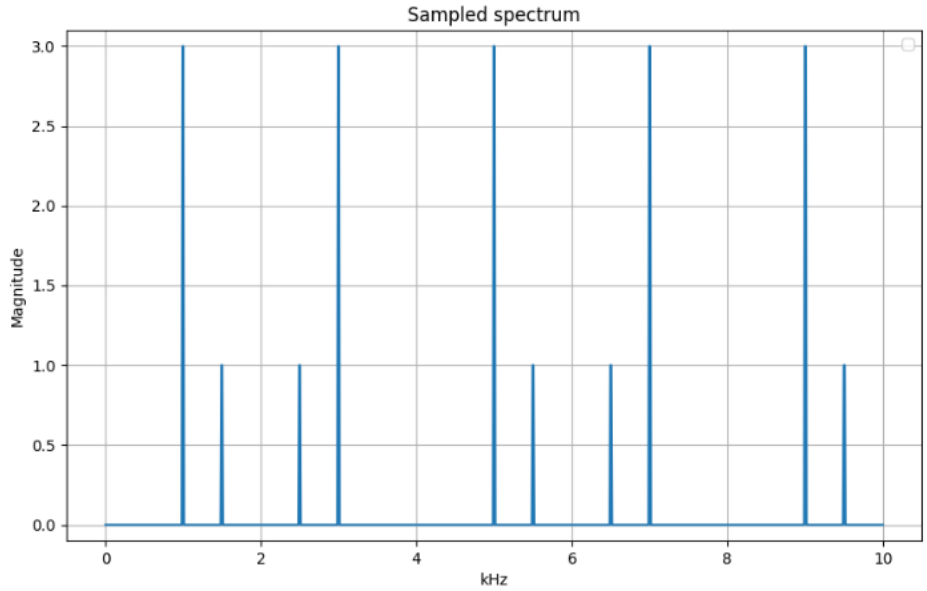
(a) The minimal sampling frequency indicated by the NSST is 6 kHz.

(b) Lokk at the example signal given by:

$$x(t) = 3\sin(2\pi \cdot 3000t) + \sin(2\pi \cdot 2500t)$$

The signal is sampled at $f_s = 4\,\text{kHz}$.

The resulting spectrum of the sampled signal $X_s(f)$ exhibits aliasing due to the fact that the signal components exceed half the sampling rate. In the frequency domain, the sampled spectrum $X_s(f)$ includes copies of the original spectrum shifted by multiples of the sampling frequency f_s.



(c) One option would be to use a low-pass filter that eliminates any frequencies over 2 kHz.

Question 5. An image is represented as a continuous function $I(x, y)$, where x and y are spatial coordinates.

(a) Explain how sampling in the spatial domain affects the frequency content of the image.

(b) If the image contains high-frequency components up to spatial frequency f_{max}, what is the minimum sampling rate (samples per unit length) required in x and y directions to prevent aliasing?

(c) Describe what happens if the image is undersampled.

Answer:

(a) Sampling in the spatial domain works very similarly to sampling in the time domain. Low sampling rates can lead to aliasing and a general loss of detail in high-frequency regions.

(b) The minimum sampling rate to prevent aliasing is given by the NSST and is equal to $2 \cdot f_{\text{max}}$. This rate should be applied in both x and y directions,

(c) If the image is undersampled, we can run into aliasing, which can lead to blocking. This can significantly degrade image quality, introducing artifacts and reducing the accuracy of any subsequent image processing or analysis tasks.

Question 6. An analog signal is quantized using a uniform quantizer with N bits per sample.

(a) If the quantizer uses $N = 8$ bits, how many quantization levels M are available?

(b) If the signal amplitude ranges from -1 V to $+1$ V, what is the quantization interval ΔV?

(c) If the number of bits is increased to $N = 10$, how does this affect the quantization interval ΔV and the Signal to Quantization Noise Ratio (SQNR)?

Answer:

(a) For an N-bit quantizer, the number of quantization levels M is given by:

$$M = 2^N \tag{17}$$

For $N = 8$ bits:

$$M = 2^8 = 256 \text{ levels} \tag{18}$$

(b) The quantization interval ΔV is the range divided by the number of levels:

$$\Delta V = \frac{2V}{256} = \frac{V}{128} \tag{19}$$

(c) When the number of bits is increased to $N = 10$:
New number of levels:

$$M_{\text{new}} = 2^{10} = 1024 \tag{20}$$

New quantization interval:

$$\Delta V_{\text{new}} = \frac{V}{512} \tag{21}$$

The quantization interval has decreased by a factor of 4.
For the SQNR, we use the formula from Lecture 3 Slide 8:

$$\text{SQNR} = 6.02N \text{ dB} \tag{22}$$

For $N = 8$:

$$\text{SQNR}_8 \approx 6.02 \cdot 8 \text{ dB} \tag{23}$$
$$\approx 48.16 \text{ dB} \tag{24}$$

For $N = 10$:

$$\text{SQNR}_{10} \approx 6.02 \cdot 10 \text{ dB} \tag{25}$$
$$\approx 60.20 \text{ dB} \tag{26}$$

The SQNR has increased by approximately 12 dB.

Question 7. A signal with a maximum amplitude $V_{\max} = 5$ V is uniformly quantized using $N = 12$ bits.

(a) Calculate the quantization interval ΔV.

(b) Assuming the quantization error is uniformly distributed between $-\Delta V/2$ and $+\Delta V/2$, calculate the mean squared quantization error σ_{qe}^2.

(c) Determine the SQNR in decibels.

Answer:

(a) The number of levels is $M = 2^{12} = 4096$. The quantization interval ΔV is:

$$\Delta V = \frac{10}{4096} V \tag{27}$$

$$\tag{28}$$

(b) The mean squared quantization error σ_{qe}^2 for a uniform distribution of error between $-\Delta V/2$ and $+\Delta V/2$ is (see slide 11 of lecture 3):

$$\sigma_{qe}^2 = \frac{(\Delta V)^2}{12} \tag{29}$$

$$\approx 4.968 \times 10^{-7} \text{ V}^2 \tag{30}$$

(c) To calculate the SQNR, we first need to determine the mean squared signal power. For a sinusoidal signal with amplitude $V_{\max}$, the mean squared value is $V_{\max}^2/2$. So:

$$\text{SQNR} \approx 6.02 \times 12 \text{ dB} \tag{31}$$

$$\approx 72.24 \text{ dB} \tag{32}$$

Question 8. Consider a sensory signal where the perception p relates to the stimulus S according to Weber's Law:

$$p = k \cdot \ln\left(\frac{S}{S_0}\right)$$

where S_0 is the threshold stimulus, and k is a constant.

(a) Explain why non-uniform quantization intervals are more suitable for such signals compared to uniform quantization.

(b) If an audio signal follows this logarithmic perception model, how would you design a quantizer to optimize the perceived audio quality?

Answer:

(a) According to Weber's Law, the perception changes logarithmically with the stimulus. This means that at low stimulus levels, small changes in the stimulus produce large changes in perception, while at high stimulus levels, larger changes in the stimulus are needed to produce the same perceptual difference. This means it may be more useful to have smaller quantization intervals for low stimulus levels and larger ones in high stimulus ranges.

(b) Design the quantizer so that the quantization intervals increase exponentially in order to match the logarithmic form of Weber's Law.

Question 9. An 8-bit grayscale image is compressed using lossy compression, resulting in a reconstructed image. The original image has pixel values $I_{\text{original}}(i,j)$, and the reconstructed image has pixel values $I_{\text{reconstructed}}(i,j)$.

(a) Define the Mean Squared Error (MSE) between the original and reconstructed images.

(b) If the peak pixel value is 255, express the Peak Signal to Noise Ratio (PSNR) in terms of MSE.

(c) Calculate the PSNR if the MSE is found to be 100.

Answer:

(a) The Mean Squared Error (MSE) between the original and reconstructed images is defined as:

$$\text{MSE} = \frac{1}{MN}\sum_{i=1}^{M}\sum_{j=1}^{N}[I_{\text{original}}(i,j) - I_{\text{reconstructed}}(i,j)]^2 \tag{33}$$

where M and N are the image dimensions (height and width in pixels).

(b) The Peak Signal to Noise Ratio (PSNR) can be written as

$$\begin{aligned}\text{PSNR} &= 10\log_{10}\left(\frac{255^2}{\text{MSE}}\right) &(34)\\ &= 20\log_{10}(255) - 10\log_{10}(\text{MSE}). &(35)\end{aligned}$$

where MAX_I is the maximum possible pixel value. For an 8-bit grayscale image, $\text{MAX}_I = 255$.

(c) Given an MSE of 100, we can calculate the PSNR via

$$\begin{aligned}\text{PSNR} &= 20\log_{10}(255) - 10\log_{10}(100) &(36)\\ &\approx 28.13 \text{ dB}. &(37)\end{aligned}$$